

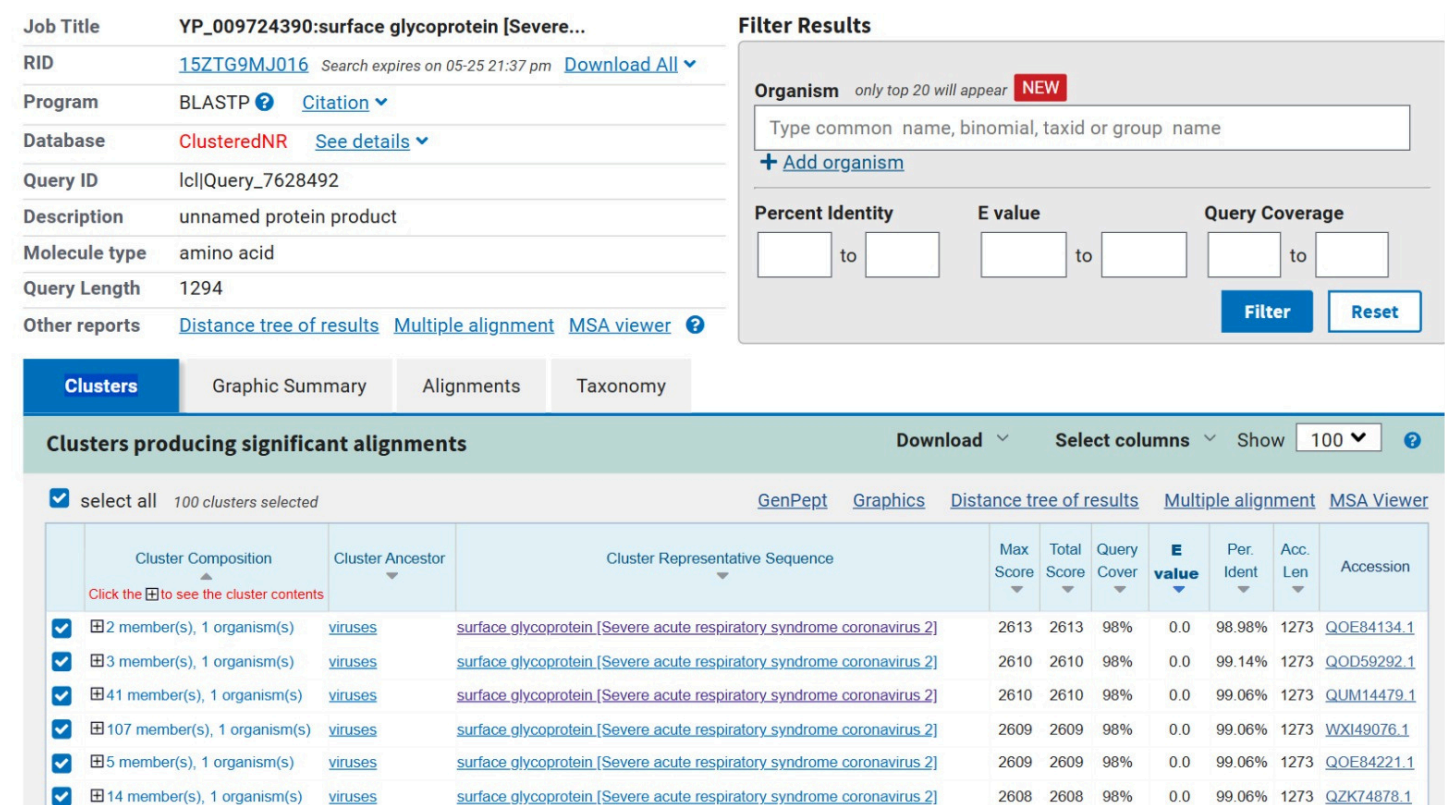
Integrated Bioinformatics Analysis of SARS-CoV-2 Spike Glycoprotein

Introduction

- SARS-CoV-2 causes COVID-19.
- Spike glycoprotein mediates host cell entry.
- Bioinformatics tools were used to analyze its sequence, structure, function, and expression.

BLAST Analysis

BLASTn
BLASTp



Objective

- Identify gene and protein sequence
- Study homology and evolution
- Analyze physicochemical properties
- Predict domains and structure
- Investigate expression and interactions

Accession number	Max score	E-value	Percent Identification	Accession Length
QOE84134.1	2613	0.0	98.98%	1273
QOD59292.1	2610	0.0	99.14%	1273
QUM14479.1	2610	0.0	99.06%	1273
WXI49076.1	2609	0.0	99.06%	1273
QOE84221.1	2609	0.0	99.06%	1273

High identity and low E-value indicate strong conservation.

Methodology

Sequence Retrieval (NCBI)

BLAST Analysis

Multiple Sequence Alignment (Clustal Omega)

Phylogenetic Analysis (MEGA)

Protein Property Analysis (ProtParam)

Domain & Motif Analysis (InterPro, CDD, MEME)

Structure Prediction (PSIPRED, AlphaFold)

Localization Prediction (DeepLoc/WoLF PSORT)

Expression Analysis (Expression Atlas)

Protein Interaction Network (STRING)

Results

Gene Identification & Sequence Retrieval

Gene : S gene

Protein: Spike glycoprotein

Organism: SARS-CoV-2 (Severe acute respiratory syndrome coronavirus 2)

Accession: YP_009724390.1

Protein Length: 1273 aa

surface glycoprotein [Severe acute respiratory syndrome coronavirus 2]

GenBank: QOE84134.1

Identical Proteins: FASTA Graphics

Go to

LOCUS QOE84134 1273 aa Linear VRL 01-OCT-2020
DEFINITION surface glycoprotein [Severe acute respiratory syndrome coronavirus 2].

ACCESSION QOE84134

VERSION QOE84134.1

DSOURCE accession [NM055367.1](#)

KEYWORDS .

SOURCE Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)

ORGANISM Severe acute respiratory syndrome coronavirus 2

VIRUSES: Riboviria; Orthornavirae; Pisuviricota; Pisoniviricetes; Nidovirales; Coronaviridae; Orthocoronavirinae; Betacoronavirus; Sarbecovirus; Betacoronavirus pandemicum.

REFERENCE 1 (residues 1 to 1273)

KEYWORDS Keyvani,H., Ranjbar,M., Soleimani,S. and Keyvani,F.

TITLE Submitted (01-OCT-2020) Virology, Iran University of Medical Sciences, Tehran Hemat Highway next to Milad Tower, Tehran, 14535, Iran

JOURNAL

COMMENT Method: conceptual translation.

FEATURES Location/Qualifiers

```
1..1273
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/coded_by="NM055367.1:1..3822"

ORIGIN
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121 nntatnviku cefqndndf lgyykhknk smesefrvy ssanctfey vsqplmdie
181 gkggrfnln efrfkndyq kisykhtpi nlrvldpfg salpdlvdp lqntfrqn
241 lllahrsytl pdgsssgeta gaayyvgyl qrtflkyld engittdpvd caldlsk
301 calksftvek ghytsnfrv qtesivfrp nitlcpfge vfnatfasv yamnrksin
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421 ynykldpdt gcviannsm ldsxvggnyv vlylfrksn lqpfndist elyagstpc
481 ngvqfnrpf plasygfat nrgngnyrv vslfella patscgkks tsklnkcn
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1261 sepvlkgt hyl
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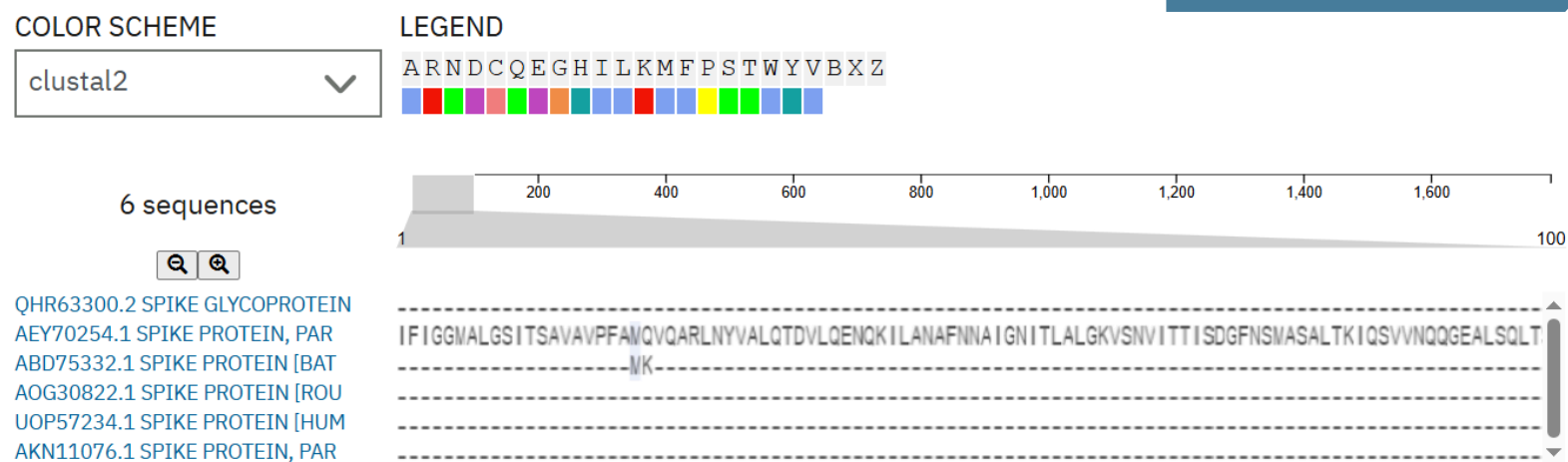
Subcellular Localization

DeepLoc

spike glycoprotein as a membrane-associated protein, consistent with its role in viral attachment and host cell entry.

Multiple Sequence Alignment & Phylogenetic Tree

Clustal Omega
MEGA



Physicochemical Analysis

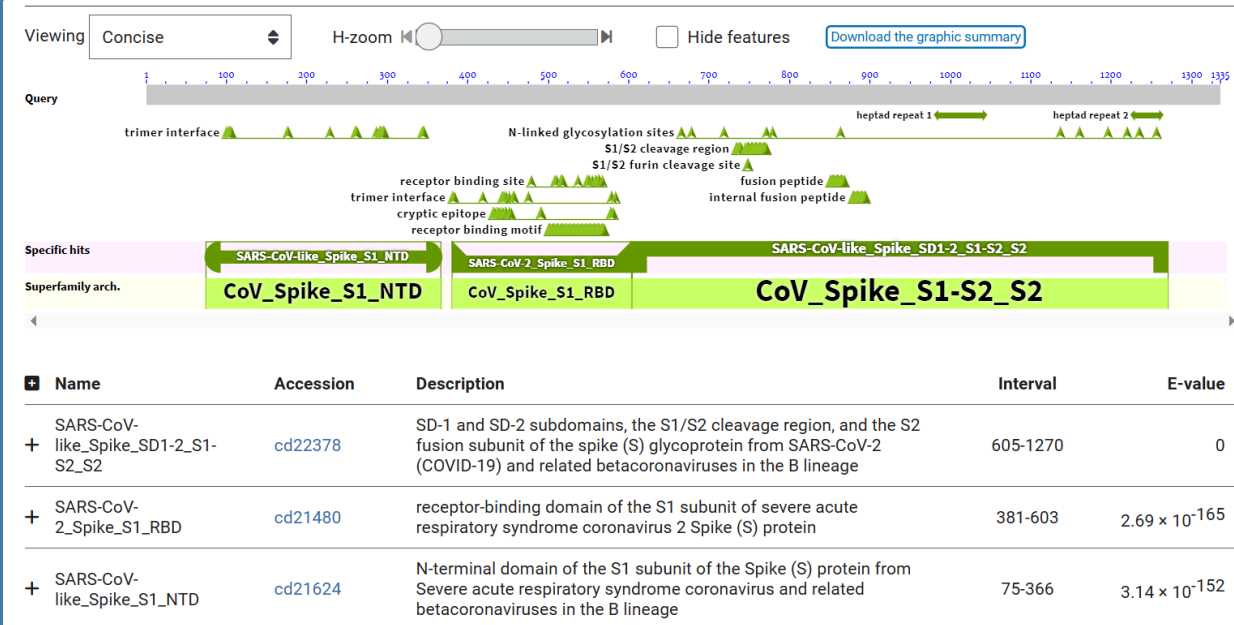
Expasy
ProtParam

Number of amino acids	1294
Theoretical pI	6.25
Molecular weight	143904.50
Aliphatic index	83.98
Grand average of hydropathicity (GRAVY)	-0.085
Instability index	33.63

The spike glycoprotein is a stable, slightly acidic, and predominantly hydrophilic protein. Its high aliphatic index indicates good thermostability

Conserved domains

NCBI CDD



Motif Analysis: MEME Suite



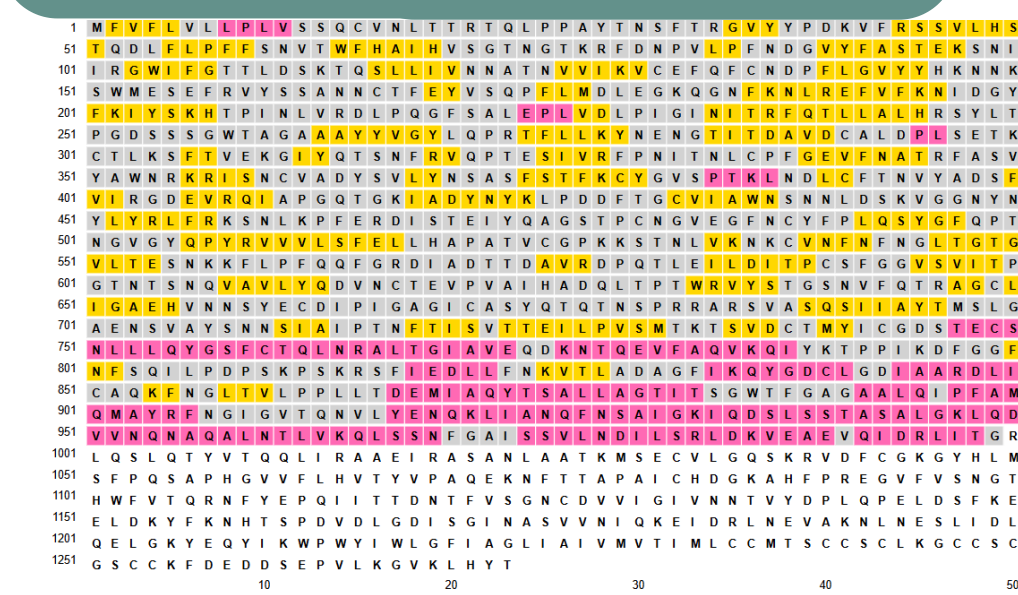
These regions are involved in

receptor binding membrane fusion, and viral entry into host cells.

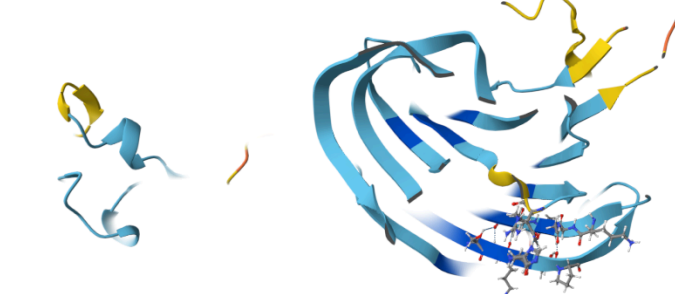
MOTIF LOCATIONS



Protein Structure Prediction:



Secondary structure prediction : PSIPRED



3D STRUCTURE: AlphaFold

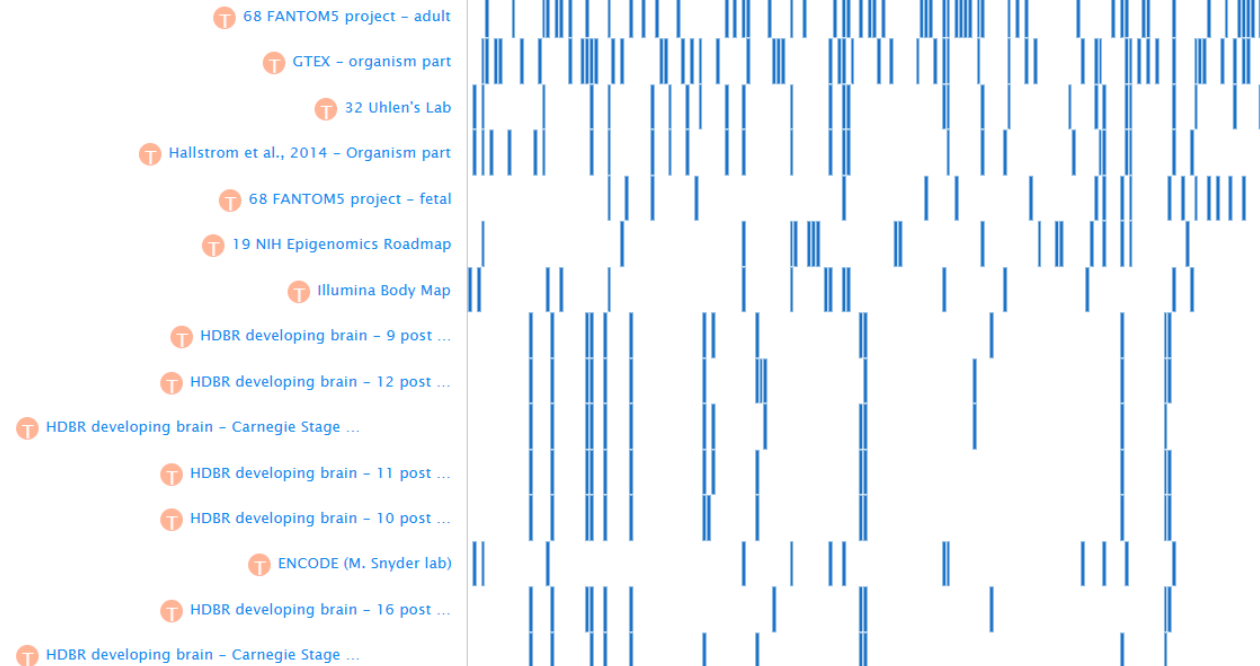
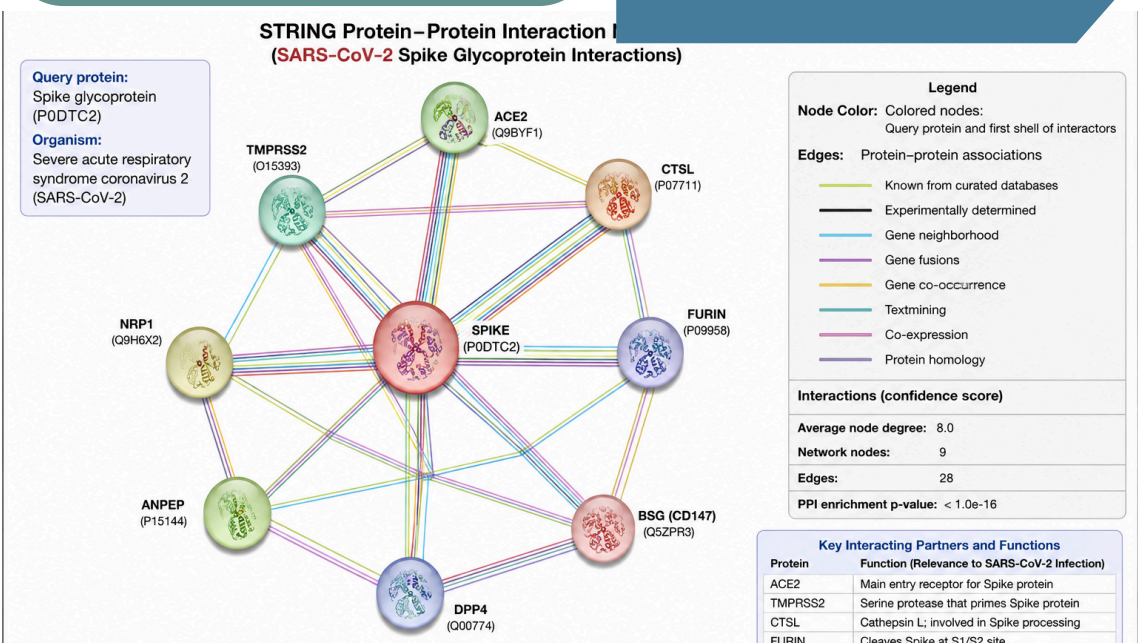
Expression Analysis

NCBI GEO

Expression patterns suggest an important role of the spike glycoprotein in coronavirus infection and host interaction.

Protein Interaction

STRING Database



Discussion

- Results support spike protein role in infection.
- Conserved domains indicate functional importance.
- Structure and interactions support viral entry mechanism.

Conclusion

Spike protein is highly conserved, Contains important receptor-binding and fusion domains. Structural analysis supports viral entry function. Potential target for vaccines and therapeutics.

REF. : NCBI, UniProt, BLAST, Clustal Omega, MEGA, ProtParam, InterPro, MEME Suite, AlphaFold, DeepLoc, Expression Atlas, STRING